

Part I

SUMMARY of Using Limits to Detect Asymptotes

Summary

February 21, 2026

Infinite Limits

$\lim_{x \rightarrow a} f(x) = \infty$ means the values of $f(x)$ grow arbitrarily large as x approaches a .

$\lim_{x \rightarrow a} f(x) = -\infty$ means the values of $|f(x)|$ grow arbitrarily large as x approaches a with $f(x)$ negative.

Limits at Infinity

$\lim_{x \rightarrow \infty} f(x) = L$ means the values of $f(x)$ becomes arbitrarily close to L by making x sufficiently large.

$\lim_{x \rightarrow -\infty} f(x) = L$ means the values of $f(x)$ becomes arbitrarily close to L by making $|x|$ sufficiently large with x negative.

Vertical and Horizontal Asymptotes:

A function f has a **vertical asymptote** at $x = a$ if at least one of the following conditions hold:

- $\lim_{x \rightarrow a} f(x) = \pm\infty$
- $\lim_{x \rightarrow a^+} f(x) = \pm\infty$
- $\lim_{x \rightarrow a^-} f(x) = \pm\infty$

A function f has a **horizontal asymptote** at $y = L$ if at least one of the following conditions hold:

- $\lim_{x \rightarrow \infty} f(x) = L$
- $\lim_{x \rightarrow -\infty} f(x) = L$

Both vertical asymptotes and horizontal asymptotes are written as the equation of a line. Vertical asymptotes are given as the equation of a vertical line, and horizontal asymptotes are given as the equation of a horizontal line.

Part II

Recitation Questions

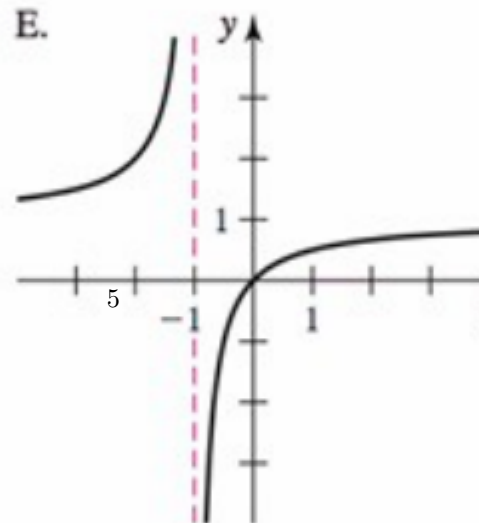
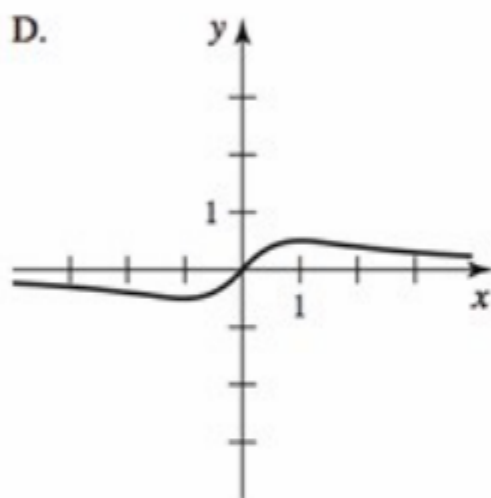
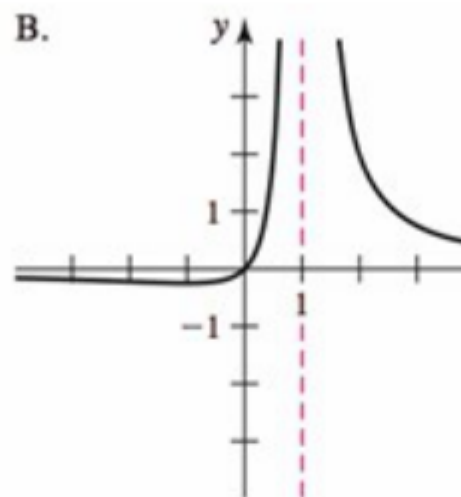
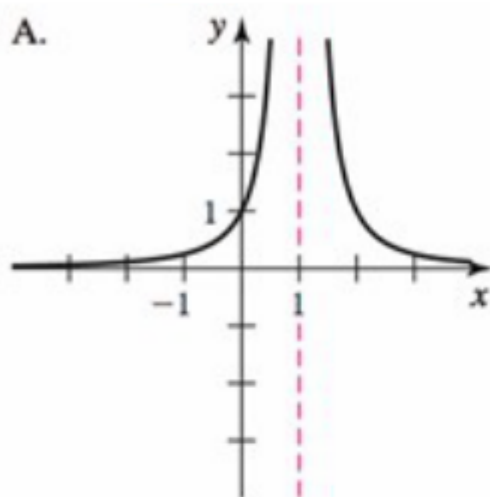
Problem 1

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Problem 1

Problem 1 Without using a graphing utility, match each graph of functions in A-F with the algebraic representation of functions in a-f:

Problem 1



- (a) The function f defined by $f(x) = \frac{x}{x^2 + 1}$.

Explanation. Since there is no real number x such that $x^2 + 1 = 0$, we have no candidates for vertical asymptotes. Hence the graph of f should not contain any vertical asymptotes. The only listed graph with no vertical asymptote is graph D.

- (b) The function g defined by $g(x) = \frac{x}{x^2 - 1}$.

Explanation. *Candidates* for vertical asymptotes:

$$x^2 - 1 = 0 \implies x = 1 \text{ or } x = -1.$$

Test of candidate $x = 1$:

$$\lim_{x \rightarrow 1^+} \frac{x}{x^2 - 1} = \lim_{x \rightarrow 1^+} \frac{x}{(x - 1)(x + 1)} = \infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator positive, the denominator also positive.

Therefore, vertical asymptote at $x = 1$.

Test of candidate $x = -1$:

$$\lim_{x \rightarrow -1^-} \frac{x}{x^2 - 1} = \lim_{x \rightarrow -1^-} \frac{x}{(x - 1)(x + 1)} = -\infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator negative, the denominator positive.

Therefore, vertical asymptote at $x = -1$.

Since $g(0) = 0$, the only listed graph that can match is graph C.

- (c) The function h defined by $h(x) = \frac{1}{x^2 - 1}$.

Explanation. *Candidates* for vertical asymptotes: $x = 1$ and $x = -1$.

Test of candidate $x = 1$:

$$\lim_{x \rightarrow 1^+} \frac{1}{x^2 - 1} = \lim_{x \rightarrow 1^+} \frac{1}{(x - 1)(x + 1)} = \infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator positive, the denominator positive.

Therefore, vertical asymptote at $x = 1$.

Problem 1

Test of candidate $x = -1$:

$$\lim_{x \rightarrow -1^-} \frac{1}{x^2 - 1} = \lim_{x \rightarrow -1^-} \frac{1}{(x - 1)(x + 1)} = \infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator positive, the denominator positive.

Therefore, vertical asymptote at $x = -1$.

Since $h(0) = -1$, the only listed graph that can match is graph F.

(d) The function a defined by $a(x) = \frac{x}{(x - 1)^2}$.

Explanation. Candidate for the vertical asymptote: $x = 1$.

Test of candidate $x = 1$:

$$\lim_{x \rightarrow 1^+} \frac{x}{(x - 1)^2} = \infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator positive, the denominator positive.

Therefore, vertical asymptote at $x = 1$.

Since $a(0) = 0$ the graph is graph B.

(e) The function s defined by $s(x) = \frac{1}{(x - 1)^2}$.

Explanation. Candidate for the vertical asymptote: $x = 1$.

Test of candidate $x = 1$:

$$\lim_{x \rightarrow 1^+} \frac{1}{(x - 1)^2} = \infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator positive, the denominator positive.

Therefore, vertical asymptote at $x = 1$.

Since $s(0) = 1$ the graph is graph A.

(f) The function r defined by $r(x) = \frac{x}{x + 1}$.

Problem 1

Explanation. Candidate for the vertical asymptote: $x = -1$.

Test of candidate $x = -1$:

$$\lim_{x \rightarrow -1^+} \frac{x}{x+1} = -\infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator negative, the denominator positive.

Therefore, vertical asymptote at $x = -1$.

Since $r(0) = 0$ the graph is graph E.

Problem 2

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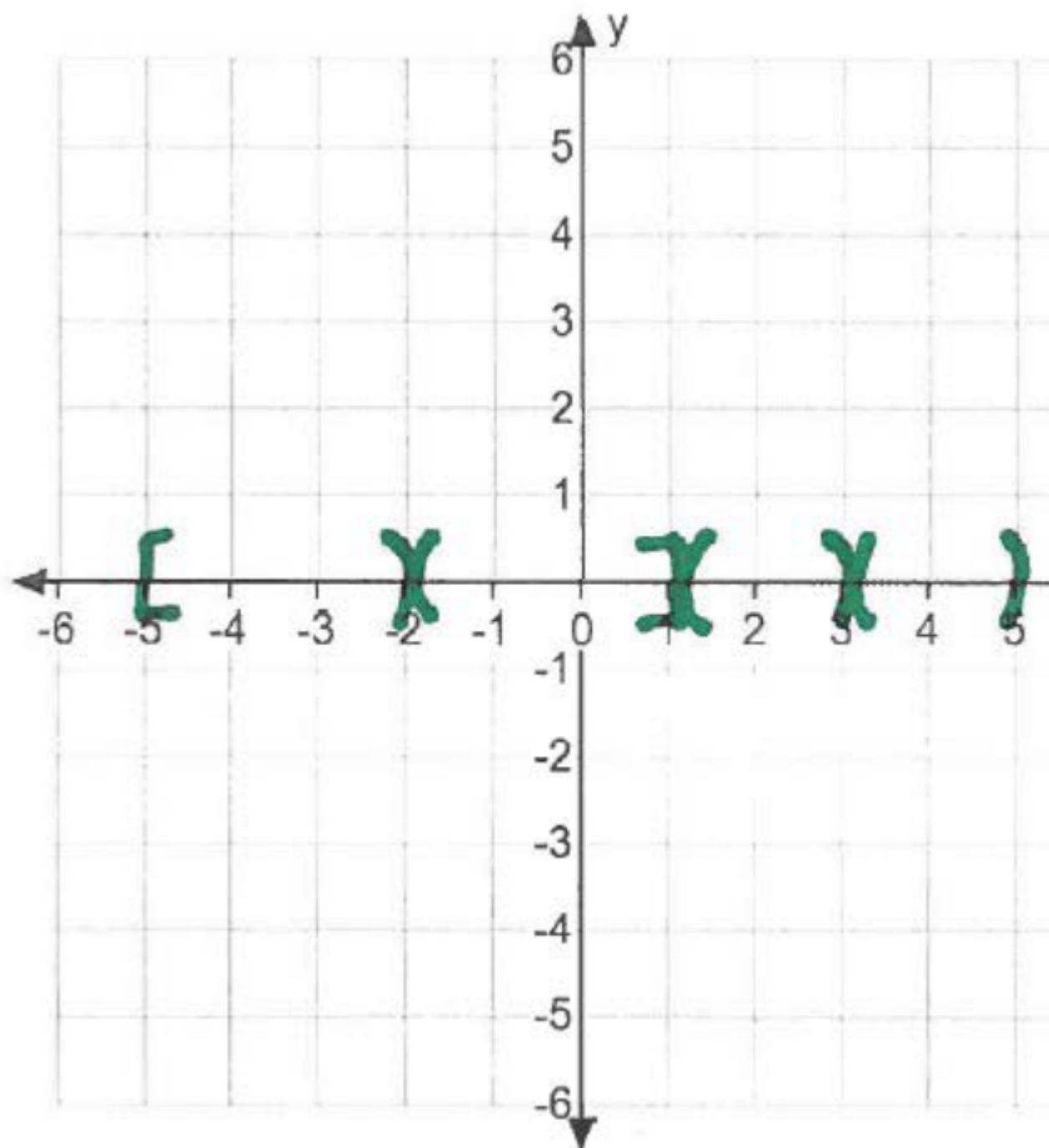
Problem 2

Problem 2 Sketch a possible graph of a function g that satisfies the following conditions:

$$\begin{array}{ll} \text{Domain: } [-5, -2) \cup (-2, 3) \cup (3, 5) & \lim_{x \rightarrow -2} g(x) = 3 \\ g(1) = 1 & \lim_{x \rightarrow 1^+} g(x) = -\infty \\ \lim_{x \rightarrow 3} g(x) = \infty & \lim_{x \rightarrow 5^-} g(x) = \infty \\ \lim_{x \rightarrow 1^-} g(x) = 1 & g(-5) = -1.8 \end{array}$$

Explanation. First we will draw our domain on the x-axis using green. We will use a bracket when a point is in the domain and curved parenthesis when it is not.

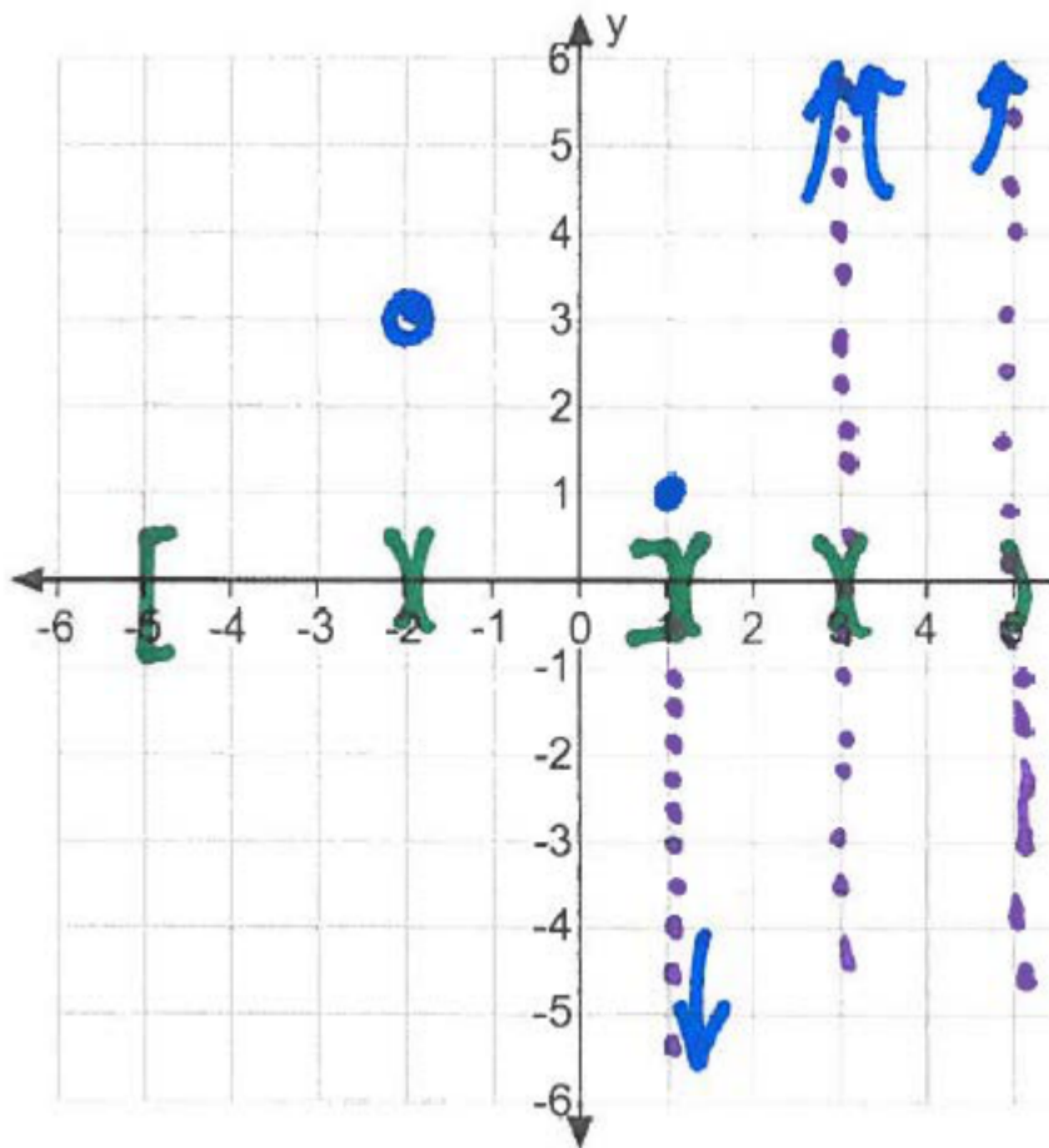
Problem 2



Problem 2

Next, we draw an open circle at $(-2, 3)$ because we know $\lim_{x \rightarrow -2} g(x) = 3$. We also draw a closed circle at $(1, 1)$ and $(-5, -1.8)$ because we know $g(1) = 1$ and $g(-5) = -1.8$. Finally we draw in our vertical asymptotes in purple as indicated by the infinite limits at $x = 1, x = 3, x = 5$. We draw the tails of the asymptotes in blue to indicate whether the function g approaches $-\infty$ or ∞ .

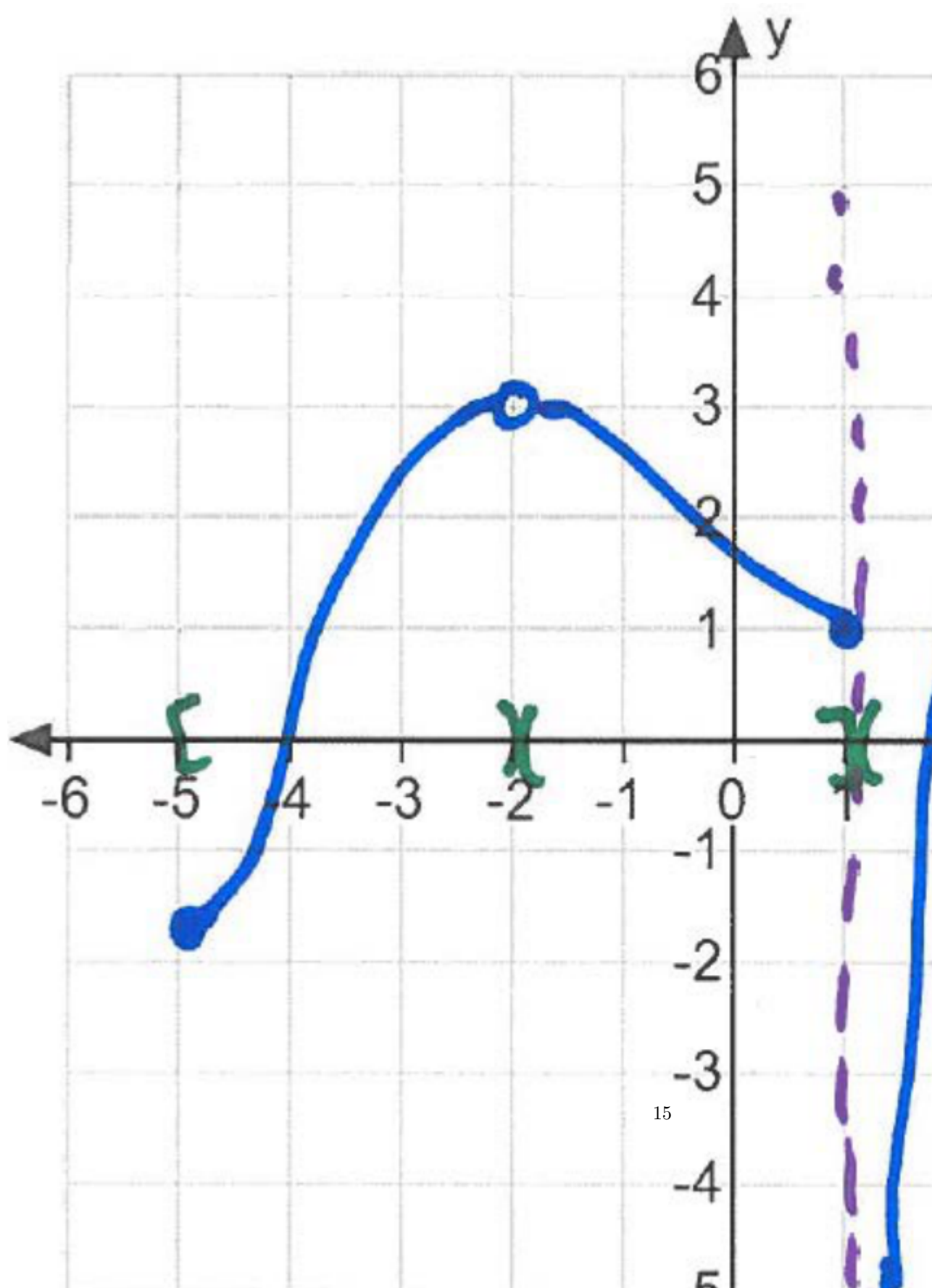
Problem 2



Problem 2

Finally, we connect our graph together.

Problem 2



Problem 3

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Problem 3 Let f be a function given by $f(x) = \ln(1 + x)$.

- (a) Find the domain of f . Write your answer in interval notation.

Explanation. $1 + x > 0 \implies x > -1 \implies \text{domain: } (-1, +\infty)$

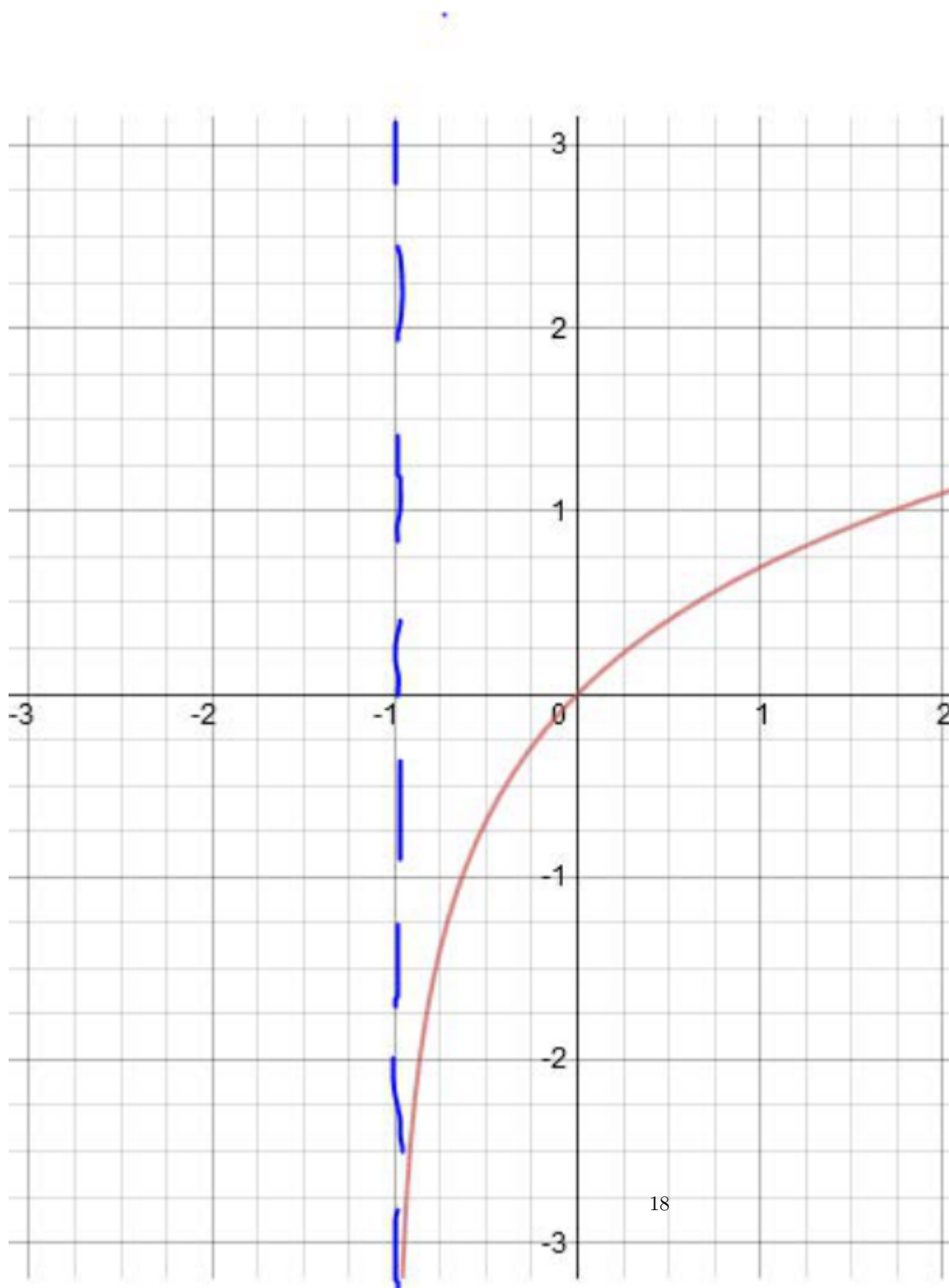
- (b) Find the vertical asymptotes of f and **EXPLAIN** and justify your answer.

Explanation. The function $f(x) = \ln(1 + x)$ has a vertical asymptote at $x = -1$ since $\lim_{x \rightarrow -1^+} \ln(1 + x) = -\infty$. (There is no Theorem/Test to reference here, since it is checking the definition of a vertical asymptote.) Alternatively, $g(x) = \ln(x)$ has a vertical asymptote at $x = 0$. f is g shifted one unit left so f will have a vertical asymptote at $x = -1$.

- (c) Sketch a graph of f

Explanation. The graph of $f(x) = \ln(1 + x)$ is show below.

Problem 3



Problem 4

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Problem 4 Let $f(x) = \frac{\ln(x)}{x-2}$.

(a) Evaluate the limit.

$$\lim_{x \rightarrow 2^-} \frac{\ln(x)}{x-2}$$

Explanation. This limit is of the form $\frac{\#}{0}$, the numerator positive, and the denominator is negative. Therefore, $\lim_{x \rightarrow 2^-} \frac{\ln x}{x-2} = -\infty$.

(b) Find the vertical asymptotes of f . **EXPLAIN** and justify your answer.

Explanation. $x = 2$ is a vertical asymptote of f since $\lim_{x \rightarrow 2^-} \frac{\ln x}{x-2} = -\infty$ calculated in part (a).

On the other hand, we know that

$$\lim_{x \rightarrow 0^+} \ln(x) = -\infty.$$

So, it follows that

$$\lim_{x \rightarrow 0^+} \frac{\ln(x)}{x-2} = \infty,$$

since the limit is of the form $\frac{\infty}{\#}$, the numerator negative, and the denominator negative. That means $x = 0$ is a vertical asymptote of f since

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{x-2} = \infty$$

Problem 5

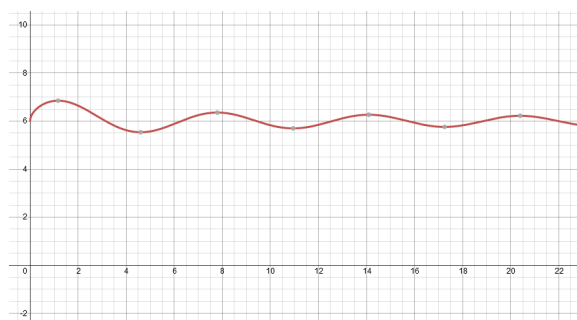
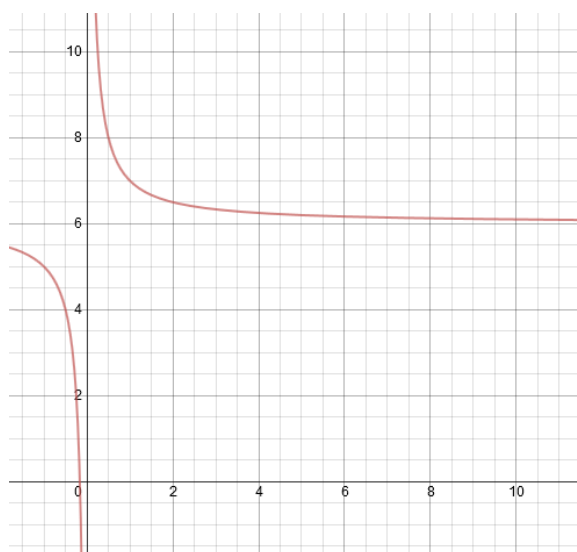
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Problem 5

Problem 5 Select the meaning of $\lim_{x \rightarrow \infty} f(x) = 6$. Support your explanation graphically.

- (a) As x becomes arbitrarily negatively large, $f(x)$ approaches 6.
- (b) As x becomes arbitrarily positively large, $f(x)$ approaches 6.
- (c) As x approaches 6, $f(x)$ becomes arbitrarily negatively large.
- (d) As x approaches 6, $f(x)$ becomes arbitrarily positively large.

Explanation. As x becomes arbitrarily positively large, $f(x)$ approaches 6. Here are two examples of what this function might look like:



Problem 6

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Problem 6

Problem 6 Evaluate the following limits.

(a) $\lim_{x \rightarrow \infty} \frac{\sqrt[3]{x^9 + 5}}{3x^3 + \sqrt{4x^6 + 1}}$

Explanation. This limit is of the form: $\frac{\infty}{\infty}$.

This will be a very detailed solution. Make sure you can justify every step

below.

$$\begin{aligned}
 \lim_{x \rightarrow \infty} \frac{\sqrt[3]{x^9 + 5}}{3x^3 + \sqrt{4x^6 + 1}} &= \lim_{x \rightarrow \infty} \frac{\sqrt[3]{x^9 + 5}}{3x^3 + \sqrt{4x^6 + 1}} \cdot \frac{\frac{1}{x^3}}{\frac{1}{x^3}} \\
 &= \lim_{x \rightarrow \infty} \frac{\frac{\sqrt[3]{x^9 + 5}}{x^3}}{\frac{3x^3 + \sqrt{4x^6 + 1}}{x^3}} \\
 &= \lim_{x \rightarrow \infty} \frac{\sqrt[3]{\frac{x^9 + 5}{x^9}}}{\frac{3x^3}{x^3} + \sqrt{\frac{4x^6 + 1}{x^6}}} \\
 &= \lim_{x \rightarrow \infty} \frac{\sqrt[3]{1 + \frac{5}{x^9}}}{3 + \sqrt{4 + \frac{1}{x^6}}} \\
 &= \frac{\lim_{x \rightarrow \infty} \sqrt[3]{1 + \frac{5}{x^9}}}{\lim_{x \rightarrow \infty} \left(3 + \sqrt{4 + \frac{1}{x^6}}\right)} \\
 &= \frac{\sqrt[3]{\lim_{x \rightarrow \infty} \left(1 + \frac{5}{x^9}\right)}}{3 + \lim_{x \rightarrow \infty} \sqrt{4 + \frac{1}{x^6}}} \\
 &= \frac{\sqrt[3]{1 + \lim_{x \rightarrow \infty} \frac{5}{x^9}}}{3 + \sqrt{\lim_{x \rightarrow \infty} \left(4 + \frac{1}{x^6}\right)}} \\
 &= \frac{\sqrt[3]{1 + 5 \lim_{x \rightarrow \infty} \left(\frac{1}{x}\right)^9}}{3 + \sqrt{4 + \lim_{x \rightarrow \infty} \frac{1}{x^6}}} \\
 &= \frac{\sqrt[3]{1 + 5 \left(\lim_{x \rightarrow \infty} \frac{1}{x}\right)^9}}{3 + \sqrt{4 + \left(\lim_{x \rightarrow \infty} \frac{1}{x}\right)^6}} \\
 &= \frac{\sqrt[3]{1 + 5(0)^9}}{3 + \sqrt{4 + (0)^6}} \\
 &= \frac{1}{3 + \sqrt{4}} \\
 &= \frac{1}{5}
 \end{aligned}$$

(b) $\lim_{x \rightarrow \infty} \frac{\sin(9x)}{5x}$

Explanation. Since $-1 \leq \sin(9x) \leq 1$ for all x , and we can divide by $5x > 0$, $\frac{-1}{5x} \leq \frac{\sin(9x)}{5x} \leq \frac{1}{5x}$.

We have $\lim_{x \rightarrow \infty} \left(-\frac{1}{5x}\right) = -\frac{1}{5} \lim_{x \rightarrow \infty} \left(\frac{1}{x}\right) = 0$ and $\lim_{x \rightarrow \infty} \frac{1}{5x} = 0$. The Squeeze theorem implies that

$$\lim_{x \rightarrow \infty} \frac{\sin(9x)}{5x} = 0$$

Problem 7

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Problem 7 The function f is defined by $f(x) = \frac{6e^x + 1}{3e^x + 5}$.

- (a) Find all vertical asymptotes of f . **EXPLAIN** and justify your answer by using appropriate limits.

Explanation. Since the exponential function, e^x , is always positive, it follows that the denominator of the function f is also always positive. Since the exponential function, e^x , is continuous on $(-\infty, \infty)$, it follows that the functions $6e^x + 1$ and $3e^x + 5$ are also continuous on $(-\infty, \infty)$. Therefore, the function f is also continuous there, since it is a quotient of two continuous functions, and the function in the denominator is never 0.

Therefore, f has no vertical asymptotes, since it is continuous on $(-\infty, \infty)$.

- (b) Find all horizontal asymptotes of f . **EXPLAIN** and justify your answer by using appropriate limits.

Explanation. Check End Behavior as $x \rightarrow \infty$:

$$\lim_{x \rightarrow \infty} \frac{6e^x + 1}{3e^x + 5} \text{ is of the form: } \frac{\infty}{\infty}$$

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{6e^x + 1}{3e^x + 5} &= \lim_{x \rightarrow \infty} \frac{6e^x + 1}{3e^x + 5} \cdot \frac{\frac{1}{e^x}}{\frac{1}{e^x}} = \\ &= \lim_{x \rightarrow \infty} \frac{6 + \frac{1}{e^x}}{3 + \frac{5}{e^x}} = \frac{6 + 0}{3 + 0} = 2 \end{aligned}$$

The line $y = 2$ is a horizontal asymptote of f because $\lim_{x \rightarrow \infty} f(x) = 2$.

Check End Behavior as $x \rightarrow -\infty$:

$$\lim_{x \rightarrow -\infty} \frac{6e^x + 1}{3e^x + 5} = \frac{0 + 1}{0 + 5} = \frac{1}{5}$$

The line $y = \frac{1}{5}$ is a horizontal asymptote of f because $\lim_{x \rightarrow -\infty} f(x) = \frac{1}{5}$.

Problem 8

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Problem 8 The function f is defined by $f(x) = \frac{\sqrt{2x^2+1}}{3x-5}$.

- (a) Find all vertical asymptotes of f . **EXPLAIN** and justify your answer by using appropriate limits.

Explanation. Candidate for vertical asymptotes: $3x - 5 = 0 \implies x = 5/3$.

Test of candidate $x = 5/3$:

$$\lim_{x \rightarrow \frac{5}{3}^+} \frac{\sqrt{2x^2+1}}{3x-5} = \infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator positive, and the denominator

$3x - 5 = 3(x - 5/3)$ is positive, since $x > 5/3$.

The line $x = \frac{5}{3}$ is a vertical asymptote of f since $\lim_{x \rightarrow \frac{5}{3}^+} f(x) = \infty$.

- (b) Find all horizontal asymptotes of f . **EXPLAIN** and justify your answer by using appropriate limits..

Explanation. Check End Behavior as $x \rightarrow \infty$:

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\sqrt{2x^2+1}}{3x-5} \text{ is of the form: } \frac{\infty}{\infty} \\ \lim_{x \rightarrow \infty} \frac{\sqrt{2x^2+1}}{3x-5} &= \lim_{x \rightarrow \infty} \frac{\sqrt{2x^2+1}}{3x-5} \cdot \frac{\frac{1}{x}}{\frac{1}{x}} \\ &= \lim_{x \rightarrow \infty} \frac{\frac{\sqrt{2x^2+1}}{\sqrt{x^2}}}{3 - \frac{5}{x}} \\ &= \lim_{x \rightarrow \infty} \frac{\sqrt{2 + \frac{1}{x^2}}}{3 - \frac{5}{x}} \\ &= \frac{\sqrt{2}}{3} \end{aligned}$$

The line $y = \frac{\sqrt{2}}{3}$ is a horizontal asymptote of f since $\lim_{x \rightarrow \infty} f(x) = \frac{\sqrt{2}}{3}$.

Check End Behavior as $x \rightarrow -\infty$:

Problem 8

$\lim_{x \rightarrow -\infty} \frac{\sqrt{2x^2+1}}{3x-5}$ is of the form: $\frac{\infty}{\infty}$

$$\begin{aligned} \lim_{x \rightarrow -\infty} \frac{\sqrt{2x^2+1}}{3x-5} &= \lim_{x \rightarrow -\infty} \frac{\sqrt{2x^2+1}}{3x-5} \cdot \frac{\frac{1}{-x}}{\frac{1}{-x}} \\ &= \lim_{x \rightarrow -\infty} \frac{\frac{\sqrt{2x^2+1}}{\sqrt{x^2}}}{-3 + \frac{5}{x}} \\ &= \lim_{x \rightarrow -\infty} \frac{\sqrt{2 + \frac{1}{x^2}}}{-3 + \frac{5}{x}} \\ &= \frac{-\sqrt{2}}{3} \end{aligned}$$

The line $y = -\frac{\sqrt{2}}{3}$ is a horizontal asymptote of f since $\lim_{x \rightarrow -\infty} f(x) = -\frac{\sqrt{2}}{3}$.

Problem 9

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Problem 9 A piecewise defined function f is given by

$$f(x) = \begin{cases} \frac{2x-3}{x-2} & \text{if } x < 2 \\ \frac{x^2-5x+6}{x^2-4} & \text{if } x > 2 \end{cases}$$

- (a) Find all vertical asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.

Explanation. Since $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{2x-3}{x-2}$, the form of the limit is $\frac{\#}{0}$.

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{2x-3}{x-2} = -\infty,$$

since the numerator positive, the denominator negative and goes to 0. Therefore, the line $x = 2$ is a vertical asymptote of f since $\lim_{x \rightarrow 2^-} f(x) = -\infty$.

Note: The line $x = -2$ is not a vertical asymptote, since for values of x near -2 , $f(x) = \frac{2x-3}{x-2}$. So, f is continuous at -2 .

- (b) Find all horizontal asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.

Explanation. Check End Behavior as $x \rightarrow \infty$:

$$\lim_{x \rightarrow \infty} \frac{x^2-5x+6}{x^2-4} \text{ is of the form: } \frac{\infty}{\infty}$$

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{x^2-5x+6}{x^2-4} &= \lim_{x \rightarrow \infty} \frac{x^2-5x+6}{x^2-4} \cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}} \\ &= \lim_{x \rightarrow \infty} \frac{1-\frac{5}{x}+\frac{6}{x^2}}{1-\frac{4}{x^2}} \\ &= 1 \end{aligned}$$

Problem 9

Therefore, the line $y = 1$ is a horizontal asymptote of f since $\lim_{x \rightarrow \infty} f(x) = 1$.

Check End Behavior as $x \rightarrow -\infty$:

$\lim_{x \rightarrow -\infty} \frac{2x-3}{x-2}$ is of the form: $\frac{\infty}{\infty}$

$$\begin{aligned}\lim_{x \rightarrow -\infty} \frac{2x-3}{x-2} &= \lim_{x \rightarrow -\infty} \frac{2x-3}{x-2} \cdot \frac{\frac{1}{x}}{\frac{1}{x}} \\ &= \lim_{x \rightarrow -\infty} \frac{2 - \frac{3}{x}}{1 - \frac{2}{x}} \\ &= 2\end{aligned}$$

Therefore, the line $y = 2$ is a horizontal asymptote of f since $\lim_{x \rightarrow -\infty} f(x) = 2$.

Problem 10

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Problem 10 For the piecewise function f defined by

$$f(x) = \begin{cases} \sin(x) & \text{if } x \leq 0 \\ \frac{x^2}{x^2 - 4} & \text{if } 0 < x < 2 \\ \frac{x^2}{x^2 + 4} & \text{if } 2 \leq x \end{cases}$$

- (a) Find all vertical asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.

Explanation. Candidates for vertical asymptote: $x^2 - 4 = 0 \implies x = -2, x = 2$ (Each piece of f is continuous at all other points).

Note: $x = -2$ is not a vertical asymptote, since $f(x) = \sin(x)$ near -2 , so $\lim_{x \rightarrow -2} f(x) = \lim_{x \rightarrow -2} \sin(x) = \sin(-2)$.

$$\lim_{x \rightarrow 2^-} \frac{x^2}{x^2 - 4} = \lim_{x \rightarrow 2^-} \frac{x^2}{(x-2)(x+2)} = -\infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator positive, and the denominator negative.

The line $x = 2$ is a vertical asymptote of f because $\lim_{x \rightarrow 2^-} f(x) = -\infty$. This is the only vertical asymptote of f .

- (b) Find all horizontal asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.

Explanation. Check End Behavior as $x \rightarrow \infty$:

$$\lim_{x \rightarrow \infty} \frac{x^2}{x^2 + 4} \text{ is of the form: } \frac{\infty}{\infty}$$

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{x^2}{x^2 + 4} &= \lim_{x \rightarrow \infty} \frac{x^2}{x^2 + 4} \cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}} \\ &= \lim_{x \rightarrow \infty} \frac{1}{1 + \frac{4}{x^2}} \\ &= 1 \end{aligned}$$

The line $y = 1$ is a horizontal asymptote of f since $\lim_{x \rightarrow \infty} f(x) = 1$.

Problem 10

Check End Behavior as $x \rightarrow -\infty$:

$\lim_{x \rightarrow -\infty} \sin(x)$ does not exist, since the function oscillates.

For example, $f(-n\pi) = \sin(-n\pi) = 0$, while $f(-(2n+1)\frac{\pi}{2}) = \sin\left(-(2n+1)\frac{\pi}{2}\right) = 1$, for all positive integers n .

There is no horizontal asymptote as $x \rightarrow -\infty$ since $\lim_{x \rightarrow -\infty} f(x)$ does not exist.

Problem 11

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Problem 11 For the function g defined by

$$g(t) = \frac{t^2 + 7t + 11}{t - 3}$$

- (a) Find all vertical asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.

Explanation. The function g is a rational function which is continuous on the intervals $(-\infty, 3)$ and $(3, \infty)$. The only candidates for a vertical asymptote is $t = 3$, where g is not defined.

Test of candidate $t = 3$:

$$\lim_{t \rightarrow 3^+} \frac{t^2 + 7t + 11}{t - 3} = \infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator positive, and the denominator also positive.

The line $t = 3$ is a vertical asymptote of g since $\lim_{t \rightarrow 3^+} g(t) = \infty$.

- (b) Find all horizontal asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.

Explanation. To find the limits, we need to divide the numerator and denominator by the highest power of t in the denominator.

$$\lim_{t \rightarrow \infty} \frac{t^2 + 7t + 11}{t - 3} = \lim_{t \rightarrow \infty} \frac{t + 7 + \frac{11}{t}}{1 - \frac{3}{t}} = \infty$$

Note: the last limit is of the form $\frac{\infty}{\#}$, the numerator positive, the denominator positive.

Therefore, g has no horizontal asymptotes as $t \rightarrow \infty$.

$$\lim_{t \rightarrow -\infty} \frac{t^2 + 7t + 11}{t - 3} = \lim_{t \rightarrow -\infty} \frac{t + 7 + \frac{11}{t}}{1 - \frac{3}{t}} = -\infty$$

Note: the last limit is of the form $\frac{\infty}{\#}$, the numerator negative, the denominator positive.

Therefore, g has no horizontal asymptotes as $t \rightarrow -\infty$ either.

Problem 12

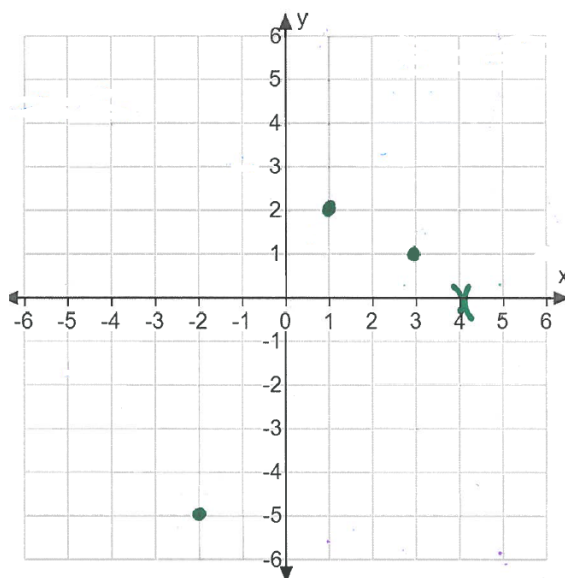
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Problem 12

Problem 12 Sketch a possible graph of a function that satisfies all of the given properties. (You do not need to find a formula for the function.)

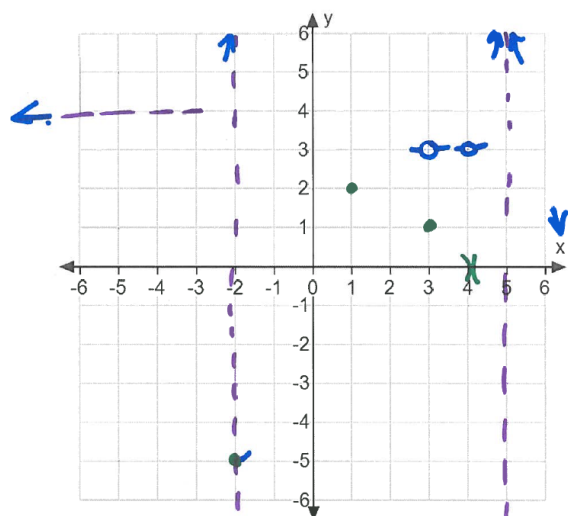
$$\begin{array}{lll} \lim_{x \rightarrow -2^-} f(x) = \infty & f(-2) = -5 & f(1) = 2 \\ \lim_{x \rightarrow \infty} f(x) = -\infty & \lim_{x \rightarrow -\infty} f(x) = 4 & \lim_{x \rightarrow 5} f(x) = \infty \\ \lim_{x \rightarrow 3} f(x) = 3 & f(3) = 1 & \\ \lim_{x \rightarrow -2^+} f(x) = -5 & f(4) \text{ is undefined} & \lim_{x \rightarrow 4} f(x) = 3 \end{array}$$

Explanation. There are many correct solutions to this problem. One possible solution can be constructed as follows: First we draw the given points. We'll also draw open brackets on the x-axis at $x = 4$ since $f(x)$ is not defined at $x = 4$ (This is seen in green.)



Then we draw the vertical and horizontal asymptotes (in purple) and arrows indicating how the graph approaches the asymptote (in blue). We also draw end-behavior as x approaches ∞ and $-\infty$ (in blue). We can also draw tails where we know limits at particular x -values. Such as $\lim_{x \rightarrow -2^+} f(x) = -5$.

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Finally we connect our graph together (seen in blue)

